

# Calculus Cheat Sheet

## Limits

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1.$$

## Derivatives

$$\frac{dx^n}{dx} = nx^{n-1}.$$

$$\frac{d|x|}{dx} = \operatorname{sgn}(x), \quad x \neq 0.$$

$$\frac{d \ln(x)}{dx} = \frac{1}{x}.$$

$$\frac{da^x}{dx} = \ln(a)a^x.$$

$$\frac{d \sin(x)}{dx} = \cos(x).$$

$$\frac{d \cos(x)}{dx} = -\sin(x).$$

$$\frac{d \tan(x)}{dx} = \sec^2(x).$$

$$\frac{d \cot(x)}{dx} = -\csc^2(x).$$

$$\frac{d \sec(x)}{dx} = \tan(x) \sec(x).$$

$$\frac{d \csc(x)}{dx} = -\cot(x) \csc(x).$$

$$\frac{d \arcsin(x)}{dx} = \frac{1}{\sqrt{1-x^2}}.$$

$$\frac{d \arccos(x)}{dx} = -\frac{1}{\sqrt{1-x^2}}.$$

$$\frac{d \arctan(x)}{dx} = \frac{1}{1+x^2}.$$

$$\frac{d \operatorname{arccot}(x)}{dx} = -\frac{1}{1+x^2}.$$

$$\frac{d \operatorname{arcsec}(x)}{dx} = \frac{1}{|x| \sqrt{x^2-1}}.$$

$$\frac{d \operatorname{arccsc}(x)}{dx} = -\frac{1}{|x| \sqrt{x^2-1}}.$$

$$\frac{d \sinh(x)}{dx} = \cosh(x).$$

$$\frac{d \cosh(x)}{dx} = \sinh(x).$$

$$\frac{d \tanh(x)}{dx} = \operatorname{sech}^2(x).$$

$$\frac{d \coth(x)}{dx} = -\operatorname{csch}^2(x).$$

$$\frac{d \operatorname{sech}(x)}{dx} = -\tanh(x) \operatorname{sech}(x).$$

$$\frac{d \operatorname{csch}(x)}{dx} = -\coth(x) \operatorname{csch}(x).$$

$$\frac{d \operatorname{arcsinh}(x)}{dx} = \frac{1}{\sqrt{x^2 + 1}}.$$

$$\frac{d \operatorname{arccosh}(x)}{dx} = \frac{1}{\sqrt{x^2 - 1}}.$$

$$\frac{d \operatorname{arctanh}(x)}{dx} = \frac{1}{1 - x^2}, \quad |x| < 1.$$

$$\frac{d \operatorname{arccoth}(x)}{dx} = \frac{1}{1 - x^2}, \quad |x| > 1.$$

$$\frac{d \operatorname{arcsech}(x)}{dx} = -\frac{1}{x\sqrt{1 - x^2}}.$$

$$\frac{d \operatorname{arccsch}(x)}{dx} = -\frac{1}{|x|\sqrt{1 + x^2}}.$$

## Integrals

$$\int x^{-1} dx = \ln|x| + C.$$

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1$$

$$\int \ln(x) dx = x \ln(x) - x + C.$$

$$\int (\ln(x))^n dx = x (\ln(x))^n - n \int (\ln(x))^{n-1} dx, \quad n \in \mathbb{N}.$$

$$\int e^{nx} dx = \frac{1}{n} e^{nx} + C, \quad n \neq 0.$$

$$\int \sin(x) dx = -\cos(x) + C.$$

$$\int \sin^2(x) dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C.$$

$$\int \sin^n(x) dx = -\frac{1}{n} \sin^{n-1}(x) \cos(x) + \frac{n-1}{n} \int \sin^{n-2}(x) dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \cos(x) dx = \sin(x) + C.$$

$$\int \cos^2(x) dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C.$$

$$\int \cos^n(x) dx = \frac{1}{n} \cos^{n-1}(x) \sin(x) + \frac{n-1}{n} \int \cos^{n-2}(x) dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \tan(x) dx = -\ln |\cos(x)| + C = \ln |\sec(x)| + C.$$

$$\int \tan^2(x) \, dx = \tan(x) - x + C.$$

$$\int \tan^n(x) \, dx = \frac{1}{n-1} \tan^{n-1}(x) - \int \tan^{n-2}(x) \, dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \cot(x) \, dx = \ln |\sin(x)| + C.$$

$$\int \cot^2(x) \, dx = -\cot(x) - x + C.$$

$$\int \cot^n(x) \, dx = -\frac{1}{n-1} \cot^{n-1}(x) - \int \cot^{n-2}(x) \, dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \sec(x) \, dx = \ln |\tan(x) + \sec(x)| + C = \ln \left| \tan\left(\frac{x}{2} + \frac{\pi}{4}\right) \right| + C.$$

$$\int \sec^2(x) \, dx = \tan(x) + C.$$

$$\int \sec^n(x) \, dx = \frac{1}{n-1} \tan(x) \sec^{n-2}(x) + \frac{n-2}{n-1} \int \sec^{n-2}(x) \, dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \csc(x) \, dx = -\ln |\cot(x) + \csc(x)| + C = -\ln \left| \cot\left(\frac{x}{2}\right) \right| + C.$$

$$\int \csc^2(x) \, dx = -\cot(x) + C.$$

$$\int \csc^n(x) \, dx = -\frac{1}{n-1} \cot(x) \csc^{n-2}(x) + \frac{n-2}{n-1} \int \csc^{n-2}(x) \, dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \arcsin(x) \, dx = x \arcsin(x) + \sqrt{1-x^2} + C.$$

$$\int \arccos(x) \, dx = x \arccos(x) - \sqrt{1-x^2} + C.$$

$$\int \arctan(x) \, dx = x \arctan(x) - \frac{1}{2} \ln(1+x^2) + C.$$

$$\int \operatorname{arccot}(x) \, dx = x \operatorname{arccot}(x) + \frac{1}{2} \ln(1+x^2) + C.$$

$$\int \operatorname{arcsec}(x) \, dx = x \operatorname{arcsec}(x) - \operatorname{sgn}(x) \ln \left| x + \sqrt{x^2-1} \right| + C, \quad |x| \geq 1.$$

$$\int \operatorname{arccsc}(x) \, dx = x \operatorname{arccsc}(x) + \operatorname{sgn}(x) \ln \left| x + \sqrt{x^2-1} \right| + C, \quad |x| \geq 1.$$

$$\int \sinh(x) \, dx = \cosh(x) + C.$$

$$\int \sinh^2(x) \, dx = -\frac{x}{2} + \frac{\sinh(2x)}{4} + C.$$

$$\int \sinh^n(x) \, dx = \frac{1}{n} \sinh^{n-1}(x) \cosh(x) - \frac{n-1}{n} \int \sinh^{n-2}(x) \, dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \cosh(x) \, dx = \sinh(x) + C.$$

$$\int \cosh^2(x) \, dx = \frac{x}{2} + \frac{\sinh(2x)}{4} + C.$$

$$\int \cosh^n(x) dx = \frac{1}{n} \cosh^{n-1}(x) \sinh(x) + \frac{n-1}{n} \int \cosh^{n-2}(x) dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \tanh(x) dx = \ln(\cosh(x)) + C.$$

$$\int \tanh^2(x) dx = x - \tanh(x) + C.$$

$$\int \tanh^n(x) dx = \frac{1}{n-1} \tanh^{n-1}(x) - \int \tanh^{n-2}(x) dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \coth(x) dx = \ln |\sinh(x)| + C.$$

$$\int \coth^2(x) dx = x - \coth(x) + C.$$

$$\int \coth^n(x) dx = -\frac{1}{n-1} \coth^{n-1}(x) - \int \coth^{n-2}(x) dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \operatorname{sech}(x) dx = 2 \arctan\left(\tanh\left(\frac{x}{2}\right)\right) + C = \arctan(\sinh(x)) + C = \arcsin(\tanh(x)) + C.$$

$$\int \operatorname{sech}^2(x) dx = \tanh(x) + C.$$

$$\int \operatorname{sech}^n(x) dx = \frac{1}{n-1} \tanh(x) \operatorname{sech}^{n-1}(x) + \frac{n-2}{n-1} \int \operatorname{sech}^{n-2}(x) dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \operatorname{csch}(x) dx = \ln \left| \tanh\left(\frac{x}{2}\right) \right| + C = \ln |\coth(x) - \operatorname{csch}(x)| + C.$$

$$\int \operatorname{csch}^2(x) dx = -\coth(x) + C.$$

$$\int \operatorname{csch}^n(x) dx = -\frac{1}{n-1} \coth(x) \operatorname{csch}^{n-1}(x) + \frac{n-2}{n-1} \int \operatorname{csch}^{n-2}(x) dx, \quad n \in \mathbb{N}_{>2}.$$

$$\int \operatorname{arcsinh}(x) dx = x \operatorname{arcsinh}(x) - \sqrt{x^2 + 1} + C.$$

$$\int \operatorname{arccosh}(x) dx = x \operatorname{arccosh}(x) - \sqrt{x^2 - 1} + C.$$

$$\int \operatorname{arctanh}(x) dx = x \operatorname{arctanh}(x) + \frac{1}{2} \ln(1 - x^2) + C.$$

$$\int \operatorname{arccoth}(x) dx = x \operatorname{arccoth}(x) + \frac{1}{2} \ln(x^2 - 1) + C.$$

$$\int \operatorname{arcsech}(x) dx = x \operatorname{arcsech}(x) + \arctan\left(\frac{\sqrt{1-x^2}}{x}\right) + C.$$

$$\int \operatorname{arccsch}(x) dx = x \operatorname{arccsch}(x) + \ln \left| x + \sqrt{1+x^2} \right| + C.$$

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad a > 0.$$

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln\left(\frac{x-a}{x+a}\right) + C = \frac{1}{a} \operatorname{arccoth}\left(\frac{x}{a}\right) + C, \quad a > 0 \wedge x^2 - a^2 > 0.$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln\left(\frac{a-x}{a+x}\right) + C = \frac{1}{a} \operatorname{arctanh}\left(\frac{x}{a}\right) + C, \quad a > 0 \wedge a^2 - x^2 > 0.$$

$$\int \frac{1}{ax+b} dx = \frac{1}{a} \ln |ax+b| + C, \quad a > 0.$$

$$\int (ax+b)^n dx = \frac{1}{a(n+1)}(ax+b)^{n+1} + C, \quad a \neq 0 \wedge n \neq -1.$$

$$\int \frac{x}{ax+b} dx = \frac{x}{a} - \frac{b}{a^2} \ln |ax+b| + C, \quad a > 0.$$

$$\int \frac{x}{(ax+b)^2} dx = \frac{1}{a^2} \ln |ax+b| + \frac{b}{a^2(ax+b)} + C, \quad a > 0.$$

$$\int x(ax+b)^n dx = \frac{a(n+1)x-b}{a^2(n+1)(n+2)}(ax+b)^{n+1} + C, \quad a \neq 0 \wedge n \notin \{-1, -2\}.$$

$$\text{Let } r = \sqrt{a^2 + x^2}.$$

$$\int r dx = \frac{1}{2} (xr + a^2 \ln(x+r)) + C.$$

$$\int xr dx = \frac{1}{3} r^3 + C.$$

$$\int \frac{1}{r} dx = \operatorname{arcsinh}\left(\frac{x}{|a|}\right) + C = \ln(x+r) + C.$$

$$\int \frac{x}{r} dx = r + C.$$

$$\text{Let } s = \sqrt{x^2 - a^2}, \quad x^2 > a^2.$$

$$\int s dx = \frac{1}{2} (xs - a^2 \ln(x+s)) + C.$$

$$\int xs dx = \frac{1}{3} s^3 + C.$$

$$\int \frac{1}{s} dx = \arcsin\left(\frac{x}{|a|}\right) + C.$$

$$\int \frac{x}{s} dx = -s + C.$$

$$\text{Let } q = \sqrt{a^2 - x^2}, \quad a^2 \geq x^2.$$

$$\int q dx = \frac{1}{2} \left( q + a^2 \arcsin\left(\frac{x}{|a|}\right) \right) + C.$$

$$\int xq dx = \frac{1}{3} q^3 + C.$$

$$\int \frac{1}{q} dx = \arcsin\left(\frac{x}{|a|}\right) + C.$$

$$\int \frac{x}{q} dx = q + C.$$

$$\text{Let } R = \sqrt{ax+b}, \quad a \neq 0.$$

$$\int R dx = \frac{2}{3a} R^3 + C.$$

$$\int x^n R dx = \frac{2}{a(2n+3)} \left( x^n R^3 - bn \int x^{n-1} R dx \right), \quad n \in \mathbb{N}.$$

$$\int \frac{1}{R} dx = \frac{2R}{a} + C.$$

$$\int \frac{x^n}{R} dx = \frac{2}{a} \left( x^n R - n \int x^{n-1} R dx \right) = \frac{2}{a(2n+1)} \left( x^n R - bn \int \frac{x^{n-1}}{R} dx \right), \quad n \in \mathbb{N}.$$

## Maclaurin Series and Taylor Series

$$x^r = \sum_{n=0}^{\infty} \binom{r}{n} a^{r-n} (x-a)^n, \quad |x-a| < |a|.$$

$$(1+x)^r = \sum_{n=0}^{\infty} \binom{r}{n} x^n, \quad r \in \mathbb{N}_0 \vee -1 < x < 1 \vee (-1 \leq x \leq 1 \wedge r > 0) \vee (x = 1 \wedge -1 < r < 0).$$

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n, \quad |x| < 1.$$

$$\frac{1}{1+x} = \frac{1}{1+a} \sum_{n=0}^{\infty} \left( \frac{a-x}{1+a} \right)^n, \quad |x-a| < |1+a|.$$

$$\sqrt{a^2 + x^2} = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} (2n)!}{4^n (n!)^2 (2n-1)} \frac{x^{2n}}{a^{2n-1}}, \quad |x| < |a|.$$

$$(1-x)^r = \sum_{n=0}^{\infty} \binom{r}{n} (-x)^n, \quad r \in \mathbb{N}_0 \vee -1 < x < 1 \vee (-1 \leq x \leq 1 \wedge r > 0) \vee (x = -1 \wedge -1 < r < 0).$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1.$$

$$\frac{1}{1-x} = \frac{1}{1-a} \sum_{n=0}^{\infty} \left( \frac{x-a}{1-a} \right)^n, \quad |x-a| < |1-a|.$$

$$\sqrt{a^2 - x^2} = - \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n-1)} \frac{x^{2n}}{a^{2n-1}}, \quad |x| < |a|.$$

$$\ln(x) = \ln(a) - \sum_{n=1}^{\infty} \frac{1}{n} \left( \frac{a-x}{a} \right)^n, \quad 0 \leq x \leq 2a \wedge a > 0.$$

$$\ln(1+x) = - \sum_{n=1}^{\infty} \frac{1}{n} (-x)^n, \quad -1 < x \leq 1.$$

$$\ln(1+x) = \ln(1+a) - \sum_{n=1}^{\infty} \frac{1}{n} \left( \frac{a-x}{1+a} \right)^n, \quad -1 < x \leq 1+2a \wedge a > -1.$$

$$\ln(1-x) = - \sum_{n=1}^{\infty} \frac{1}{n} x^n, \quad -1 \leq x < 1.$$

$$\ln(1-x) = \ln(1-a) - \sum_{n=1}^{\infty} \frac{1}{n} \left( \frac{x-a}{1-a} \right)^n, \quad 2a-1 \leq x < 1 \wedge a > 1.$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

$$e^x = e^a \sum_{n=0}^{\infty} \frac{(x-a)^n}{n!}.$$

$$b^x = \sum_{n=0}^{\infty} \frac{(\ln b)^n}{n!} x^n, \quad b > 0.$$

$$b^x = b^a \sum_{n=0}^{\infty} \frac{(\ln b)^n}{n!} (x - a)^n, \quad b > 0.$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}.$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{\sin\left(a + \frac{n\pi}{2}\right)}{n!} (x - a)^n.$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}.$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{\cos\left(a + \frac{n\pi}{2}\right)}{n!} (x - a)^n.$$

$$\arcsin(x) = \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}, \quad |x| \leq 1.$$

$$\arccos(x) = \frac{\pi}{2} - \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}, \quad |x| \leq 1.$$

$$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}, \quad |x| \leq 1.$$

$$\operatorname{arccot}(x) = \frac{\pi}{2} - \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}, \quad |x| \leq 1.$$

$$\sinh(x) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1}.$$

$$\sinh(x) = \sum_{n=0}^{\infty} \frac{((n+1) \bmod 2) \sinh a + (n \bmod 2) \cosh a}{n!} (x - a)^n.$$

$$\cosh(x) = \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n}.$$

$$\cosh(x) = \sum_{n=0}^{\infty} \frac{((n+1) \bmod 2) \cosh a + (n \bmod 2) \sinh a}{n!} (x - a)^n.$$

## Multivariate Calculus

$$\int_C \nabla \mathbf{F} \, d\mathbf{r} = \mathbf{F}(\mathbf{r}(b)) - \mathbf{F}(\mathbf{r}(a)), \quad C : \mathbf{r}(t), \quad t \in [a, b].$$

$$\int_V \nabla \cdot \mathbf{F} \, dV = \oint_S \mathbf{F} \cdot d\mathbf{S}, \quad S = \partial V.$$

$$\iint_S \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \, dS = \oint_{\partial S} P \, dx + Q \, dy.$$

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\ell} \mathbf{F} \cdot d\boldsymbol{\ell}, \quad \ell = \partial S.$$